

Ubuntu screen Commands for Running a Python API

A practical cheat sheet for keeping Python, Flask, or FastAPI services running after you disconnect from SSH.

Important: screen keeps a process running after terminal/SSH disconnects. It does not automatically restart your API after a server reboot or process crash. For true production 24/7 behavior, use systemd or a process manager.

1. Install and verify screen

Install screen

```
sudo apt update
sudo apt install screen -y
screen --version
```

2. Basic session workflow

Action	Command / Keys
Start named session	<code>screen -S python_api</code>
Detach session	<code>Ctrl + A</code> , then <code>D</code>
List sessions	<code>screen -ls</code>
Reattach by name	<code>screen -r python_api</code>
Reattach by ID	<code>screen -r 12345</code>
Force reattach	<code>screen -d -r python_api</code>
Exit inside screen	<code>exit</code>
Kill session from outside	<code>screen -S python_api -X quit</code>
Clean dead sessions	<code>screen -wipe</code>

3. Run a Python API inside screen

Normal Python API

```
cd /path/to/your/project
source venv/bin/activate
screen -S python_api
python3 app.py
```

FastAPI / Uvicorn

```
cd /path/to/your/project
source venv/bin/activate
screen -S python_api
uvicorn main:app --host 0.0.0.0 --port 8000
```

Flask

```
cd /path/to/your/project
source venv/bin/activate
screen -S flask_api
flask run --host=0.0.0.0 --port=5000
```

After starting the server, detach with **Ctrl + A**, then **D**. The process continues running in the screen session.

4. Start directly in the background

Use these when you want to create a detached screen session and start the API in one command.

Detached normal Python command

```
screen -dmS python_api bash -c "cd /path/to/project && source venv/bin/activate && python3
app.py"
```

Detached FastAPI / Uvicorn command

```
screen -dmS python_api bash -c "cd /path/to/project && source venv/bin/activate && uvicorn
main:app --host 0.0.0.0 --port 8000"
```

Check and reconnect

```
screen -ls
screen -r python_api
```

5. Logging commands

Run Python API and save logs

```
python3 app.py >> api.log 2>&1
```

Run FastAPI and save logs

```
uvicorn main:app --host 0.0.0.0 --port 8000 >> api.log 2>&1
```

Watch API logs

```
tail -f api.log
```

Start screen with screen-level logging

```
screen -L -Logfile api_screen.log -S python_api
```

Action	Command / Keys
Toggle screen logging	Ctrl + A, then H
Default screen log file	screenlog.0

6. Window controls inside screen

Action	Command / Keys
Create new window	Ctrl + A, then C
Next window	Ctrl + A, then N
Previous window	Ctrl + A, then P
Show window list	Ctrl + A, then "
Rename current window	Ctrl + A, then A
Kill current window	Ctrl + A, then K, then y
Show help	Ctrl + A, then ?

Example layout: window 1 runs your API, window 2 watches logs, and window 3 is available for Git or shell commands.

7. Scrollback and copy mode

Action	Command / Keys
Enter scroll mode	Ctrl + A, then Esc
Scroll	Up / Down / PageUp / PageDown
Exit scroll mode	Esc or q

8. Split panes inside screen

Action	Command / Keys
Horizontal split	Ctrl + A, then S
Vertical split	Ctrl + A, then
Move to next pane	Ctrl + A, then Tab
Close current pane	Ctrl + A, then X

A window is a full terminal inside screen. A pane is a split view inside the current screen display.

9. Send commands into an existing screen session

Send a command to a screen session

```
screen -S python_api -X stuff "ls -la$(printf '\r')"
```

This is useful for automation, but use carefully. For stopping an API, it is usually clearer to reattach and press Ctrl + C.

10. Port check and port cleanup commands

Use these before starting your API, especially when ports like 5000, 8000, 8080, 80, or 443 are already in use.

Show all listening TCP/UDP ports

```
sudo ss -tulpn
```

Check one specific port

```
sudo ss -tulpn | grep :8000
sudo lsof -i :8000
```

Free a port by killing the process

```
sudo kill PID
sudo kill -9 PID
```

One-command kill for a TCP port

```
sudo fuser -k 8000/tcp
```

Check common web/API ports

```
sudo ss -tulpn | grep -E ':5000|:8000|:8080|:3000|:80|:443'
```

Stop common services occupying ports 80 or 443

```
sudo systemctl stop nginx
sudo systemctl stop apache2
```

11. Recommended Python API workflow

Start API

```
cd /path/to/your/project
source venv/bin/activate
screen -S python_api
python3 app.py >> api.log 2>&1
```

Detach and leave it running

```
Ctrl + A, then D
```

Check later

```
screen -ls
screen -r python_api
tail -f api.log
```

Stop API and close screen

```
screen -r python_api
Ctrl + C
exit
```

12. Troubleshooting

Action	Command / Keys
screen not found	sudo apt install screen -y
Session already attached	screen -d -r python_api
No session found	screen -ls

Action	Command / Keys
Remove dead sessions	<code>screen -wipe</code>
Port already in use	<code>sudo lsof -i :8000</code>
Server still rejects connection	Check firewall/security group and bind to 0.0.0.0
Need restart after reboot	Use systemd instead of only screen

13. Minimal cheat sheet

Most-used commands

```
# Start
screen -S python_api
python3 app.py

# Detach
Ctrl + A, then D

# List
screen -ls

# Reconnect
screen -r python_api

# Force reconnect
screen -d -r python_api

# Kill session
screen -S python_api -X quit
```